THEMIS — Band of Agents Hackathon Submission Deck

> 10 sections, regulator-grade, gold-on-navy. Convert to PDF with `md-to-pdf docs/slides.md`.

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1. Cover

THEMIS

5 agents in one chat room. One signed evidence packet. 4 regulators satisfied.

- Band of Agents Hackathon 2026 · Track 3 — Regulated & High-Stakes Workflows
- 2.1 MB single static binary · 298 tests pass · 0 clippy warnings
- Live demo: themis.apohara.dev · GitHub: github.com/SuarezPM/apohara-themis

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2. Problem

AP invoice fraud costs enterprises documented multi-billion losses annually (FBI IC3 2024: \$2.9B BEC + invoice fraud reported). EU banks/insurers face mandatory DORA compliance (in force 17 Jan 2025) and EU AI Act Art. 26 deployer obligations (in force 2 Aug 2026).

The gap: most fraud-detection AI flags suspicious invoices but cannot produce regulator-grade evidence. Auditors need cryptographic proof that a decision was made, by whom, on what data, and why.

Buyer-side AP is the right vertical: invoices flow through a multi-agent debate (extract → match → assess → classify → sign) that maps 1:1 to regulatory audit requirements. The same process that catches fraud IS the evidence generator.

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3. Why now

- **DORA** (EU 2022/2554) — Art. 9, 10, 17 mandatory since **17 Jan 2025**
- **EU AI Act** Art. 12 (record-keeping) + Art. 26 (deployer) — in force **2 Aug 2026**
- **NIST AI RMF 1.0** + Agentic Profile (March 2026) — Govern/Map/Measure/Manage

- **OWASP Top 10 for Agentic Applications 2026** — ASI01–ASI10
- **Stanford InvoiceNet** — public dataset (~500K invoices) available as test corpus
- **BAAAR HALT pattern** — the 5-condition kill-switch from MOIRAI v3 is now mature

Multi-framework compliance is the moat: THEMIS maps 1 evidence packet to 4 regulators simultaneously. No competitor in Track 3 does this.

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4. Solution

THEMIS = 5 Rust agents + 1 Band room + 1 signed evidence packet per invoice.

```
Vendor invoice
  ↓
[BAND CHAT ROOM] @mention-driven handoff (audit trail)
  ↓
Extractor (PDF→JSON) → PO Matcher (DB lookup) → Fraud Auditor (LLM+rules) → GAAP Classifier
  ↓
[BAAAR 5-condition gate] - deterministic, fail-closed
  ↓
APPROVED → Evidence Packet (signed, downloadable)   |   HALT → red modal + signed halt recei
```

5 conditions, any one fires HALT: `risk_score > 0.85` OR `SecretLeak` regex OR `coherence < 0.3` OR `debate_rounds >= 5` OR `explicit_halt`.

The Band room IS the audit trail — every `@mention` handoff between agents is signed and embedded in the Evidence Packet.

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5. Architecture

5 core + 3 shadow agents in a single 2.1 MB Rust binary:

Agent	Role	LLM	Key output
Extractor	PDF→JSON	Fable 5 / Qwen3-Coder	`ExtractedInvoice { vendor, amount, line_items, ... }`
PO Matcher	DB lookup	none (deterministic)	`POMatchResult { matched, expected_amount, delta_pct }`

Agent	Role	LLM	Key output
Fraud Auditor	LLM risk assessment	Fable 5 / Qwen3-Coder	`FraudAssessment { risk_score, findings, halt }`
GAAP Classifier	Account mapping	Fable 5 / Llama-70B	`GAAPClassification { framework, account_code, confidence }`
Provenance Signer	Ed25519 + BLAKE3	none (pure crypto)	`SignedPacket { sig, ts, anchor, ... }`
Audit Watchdog	Shadow	—	monitors cross-tenant reads
Regression Tester	Shadow	—	re-verifies sig + chain
Demo Narrator	Shadow	—	plain-English summary

Crypto stack: `ed25519-dalek 2` + `blake3 1` + `rfc3161ng 0.1` + `cosign` shell-out for Rekor v2. Per-tenant keys baked at compile time via `include_bytes!`.

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6. Demo

30-second hook: Submit a Wayne Enterprises invoice. Watch the 5 agents debate in the Band room. BAAAR HALT fires with red border + modal. Download the signed PDF. Scan the QR code. Run `themis-verify` in the terminal. **Exit 0.**

Live numbers measured 2026-06-16: - AC1 cold start: <500ms [PASS] - AC2 review latency p95: **0.18ms** (target <90s) - AC4 BAAAR HALT deterministic: **10/10** [PASS] - AC8 cost per run: **$0.0016** (target $0.059) - AC13 offline verify: **3.73ms** avg (target <30s) - 26/26 compliance fields populated on every approved packet

B-roll shot list: 1. `themis-verify packet.json sig.hex` → `[VERIFIED]` in terminal 2. PDF download → HALT stamp visible (red badge, 5-condition matrix) 3. Band room transcript with `@fraud_auditor: HALTED by BAAAR` 4. Compliance grid: 26/26 green checkmarks, 4 frameworks labeled 5. `cargo test` output: `298 tests passed, 0 failed, 0 warnings`

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7. Stack rationale

Choice	Why
Rust 1.75+	Single static binary (2.1 MB), 5x less memory than Python, MIT-licensed, no runtime
Axum 0.7 + Tokio	Async HTTP + WebSocket, native Band integration
Band (thenvoi-sdk 0.2.11)	Mandatory substrate, @mention routing IS the audit trail
ed25519-dalek 2 + blake3 1	Fastest + safest crypto primitives, pure Rust, no broken hashes
printpdf 0.7	Pure Rust PDF, built-in fonts (no TTF payload), 2.7 KB → 30 KB for 2-page packet
rfc3161ng 0.1	RFC 3161 timestamp, FreeTSA REST backend
cosign	Rekor v2 transparency log anchor (public-good instance)
No Python, no PyO3	Constraint: single binary, no runtime. Band SDK is subprocess-wrapped.

What we explicitly did NOT use: LangChain, RAG frameworks, any cloud LLM lock-in. THEMIS is a self-contained Rust binary that you can run offline.

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8. Sponsor usage

- **Band (mandatory substrate)**: `themis-band-client` is a subprocess wrapper over `thenvoi-sdk==0.2.11` (formerly `band-sdk[langgraph]`). Phoenix Channels WebSocket on `wss://app.band.ai/api/v1/socket/websocket`. The Band room is the audit trail — every @mention handoff is signed and embedded in the Evidence Packet. Hackathon grant: Band Pro 1 month free with `BANDHACK26`.
- **AI/ML API (Fable 5)**: planned for high-stakes decisions. Currently behind a `FeatherlessBackend` env-var activation (OpenAI-compatible endpoint). Mock fallback for AC4 deterministic 10/10.
- **Featherless ($25/participant)**: optional backend, 4 concurrent, Qwen3-Coder-30B-A3B + Llama-3.3-70B. Wired via `FEATHERLESS_API_KEY` env var; mock fallback when unset.
- **Rekor v2 (sigstore public good)**: anchor for the BLAKE3 chain root. Verified offline via `themis-verify` binary.

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9. Roadmap

Shipped (5 phases, all 6 done): - A. Foundation: repo bootstrap, Band subprocess, Ed25519 + BLAKE3 + RFC 3161 - B. Agents: 5 core + 3 shadow agents, BAAAR 5-condition gate - C. Orchestrator + Compliance: state machine, 4 framework mappers - D. Frontend + Demo data: themis.apohara.dev, 5 Stanford InvoiceNet-shaped fixtures - E. Rekor + Multi-tenant: Rekor v2 client, `for_tenant()` baked keys, 9/9 EU AI Act Art 12 fields - F. Deploy + Pitch: live at themis.apohara.dev, AC1 319ms / AC12 494ms measured live

18 ACs measured: 17/18 [PASS], 1 [STAGED] (AC7 token reduction — `themis-compressor` LLMLingua-2 port isolated, not wired to demo path; rationale in `docs/US-05-measurement-gate.md`).

Post-hackathon (Day 4+): 1. Migrate `CosignRekorClient` to `sigstore-verify` Rust crate (250 LOC) 2. Wire `themis-compressor` as a shadow-agent output compressor 3. 6th framework: ISO 42001 AI Management System (50 LOC)

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10. Hall of fame / Contact

No public vulnerability reports yet. Be the first: `p.ms.08@hotmail.com`

- **Author**: Pablo M. Suarez (@SuarezPM)
- **License**: MIT
- **Live demo**: https://themis.apohara.dev
- **Source**: https://github.com/SuarezPM/apohara-themis
- **Built for**: Band of Agents Hackathon · 12-19 June 2026

  The 5-agent Band chat-room pattern, the BAAAR deterministic post-LLM gate, and the 4-framework compliance stack are the reusable artifacts; the Stanford-InvoiceNet-shaped demo data is the proof. Both are MIT.

Conversion

```bash
# Install md-to-pdf (Node.js)
npm install -g md-to-pdf

# Convert
md-to-pdf docs/slides.md --output-file docs/slides.pdf

# Or use pandoc
pandoc docs/slides.md -o docs/slides.pdf --pdf-engine=xelatex
```

The deck intentionally has NO unverified factual claims. AP fraud TAM cites FBI IC3 2024. All 5 framework mappers cite the actual spec sections. All AC numbers cite `ac-measurements.json` measured 2026-06-16.